HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and modified electrical schematics and layouts for moderate-scope assemblies under general supervision, ensuring adherence to specifications.
- Coordinated procurement and assembly of electrical components, identifying sources for critical parts to resolve technical issues and ensure equipment performance.
- Troubleshot moderately complex electrical problems and implemented solutions, contributing to the development of electrical product specifications.
- Performed basic engineering analysis and generated test reports, including alpha and gamma documentation for product releases.
- Analyzed and debugged complex electrical issues across multiple hardware and software components, reducing resolution time by approximately 15%.
- Developed and implemented concepts for electrical solutions to address product issues of moderate difficulty and scope.
- Containerized electrical design toolchains and simulation environments using **Docker** for consistent deployment across development and testing stages on **AWS EC2**.

SKILLS

Electrical Engineering – Schematics, Layouts, Assemblies, Test Reports, Product Specifications, Troubleshooting, Documentation, Procurement

Software & Tools – **Docker**, **AWS**, **EC2**, **CloudWatch**, **Linux**, **Shell Scripting**, **Jira**, **Confluence**, **Git**, **Jenkins**

Programming & Scripting – **Python**, **Java**, **SQL**

Data Analysis & Visualization – **Power BI**, **Tableau**, **DAX**, **Power Query**, **R**, **pandas**, **NumPy**

Cloud Platforms – **AWS** (EC2, S3, Lambda, CloudWatch)

Databases – **PostgreSQL**, **MySQL**, **Oracle**

PROJECTS

Automated Electrical Design Validation System

- Developed a **Python** script to automate the validation of electrical schematics against predefined design rules, reducing manual review time by 40%.
- Integrated **Jira** and **Confluence** for issue tracking and documentation, creating automated tickets for design deviations identified during validation.
- Implemented shell scripts on **Linux** to manage build environments and deploy validation tools to **AWS EC2** instances.

Embedded Systems Performance Analysis Dashboard

- Engineered data extraction pipelines using **Python** and **SQL** from embedded device logs stored in **AWS S3**.
- Designed and built interactive dashboards in **Power BI** to visualize real-time performance metrics and identify bottlenecks in embedded systems.
- Applied statistical methods using **R** and **pandas** to analyze trends in device error rates, correlating them with specific hardware configurations.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)